

Automated Customer Reviews Classifier

NLP · Sentiment Analysis · Category Clustering · Product Scoring · Groq LLM

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Type	Project Report
Stack	Python · HuggingFace · Streamlit · Groq · Google Colab

Project Overview

The **Automated Customer Reviews Classifier** transforms raw Amazon product review CSVs into structured marketing intelligence through a four-stage NLP pipeline exposed via a single-page Streamlit app.

34,626+

Reviews Analysed

4

Pipeline Stages

4

Product Groups

3

App Views

01

Sentiment Classification

02

Category Clustering

03

Product Scoring

04

AI Summarisation

Problem & Motivation

Problem	Description
Scale	Thousands of reviews per product — manual reading is impractical.
Fragmentation	Raw category strings are inconsistent across datasets.
Weak signal	Star ratings miss nuance — a 3-star review can praise or criticise.
Marketing gap	Data science outputs need translating into actionable business language.

Data Source

Datafiniti Consumer Reviews of Amazon Products (Kaggle) — 34,626 reviews, 24 columns. Key fields: *reviews.text*, *reviews.rating* (1–5), *categories*, *name*, *reviews.doRecommend*.

Preprocessing: drop irrelevant columns, remove rows missing text or rating, fill nulls, normalise types, engineer *reviews_sentiment* ground-truth label from star rating (4–5→positive, 3→neutral, 1–2→negative).

Stage 1 — Sentiment Classification

Model: *distilbert-base-uncased-finetuned-sst-2-english*

- Combines review title + body for full context
- GPU batch inference via Colab/FastAPI backend
- Output: sentiment label + confidence score per review
- Ground truth: star-rating derived labels for evaluation

Performance: Strong on positive class (72%). Neutral hardest — genuine ambiguity in 3-star reviews. Average confidence >0.85.

Sentiment	Share	Notes
Positive	72%	Dominant class — most Amazon reviews are positive
Neutral	16%	Smallest class — 3-star reviews genuinely ambiguous
Negative	12%	Well separated from positive class

Stage 2 — Category Clustering

Consolidates inconsistent product category strings into 4 clean groups using semantic embeddings and hierarchical clustering.

- Normalise: lowercase, strip punctuation, remove domain stop words
- Encode with sentence-transformers/all-MiniLM-L6-v2 (384-dim)
- Cluster with Ward linkage hierarchical clustering (scipy)
- Map original category strings to cluster labels

Cluster	Match Criteria	Examples
Kindle E-Readers	Contains 'kindle' or 'e-reader'	Kindle Paperwhite, Kindle Basic
Fire Tablets	Contains 'tablet' or 'fire'	Fire HD 8, Fire HD 10
Smart Assistants	Contains 'alexa', 'echo', or 'tv'	Echo Dot, Echo Show
Unbranded	All other categories	Generic e-reader listings

Stage 3 — Product Scoring

Ranks products using a weighted composite score from review signals, with Bayesian averaging to penalise products with few reviews.

- Signals: rating, review length, sentiment, doRecommend, numHelpful
- Weights: learned via LinearRegression against star rating ($R^2 \sim 0.6\text{--}0.7$)
- Bayesian rank: $(C \cdot m + n \cdot \text{score}) / (C + n)$ — penalises low review counts

Product	Score (/100)
Kindle Paperwhite	87
Echo Dot (3rd)	83
Fire HD 8	78
Kindle Basic	75
Echo Show	71

Scores above are illustrative.

Stage 4 — AI Summarisation (Groq)

Generates marketing-ready summaries using **Groq mixtral-8x7b-32768** (temp 0.7, 1000 tokens). Configurable by summary type, product name filter, and optional signals (price, date range, sentiment).

Summary Type	Focus
Overview	General product and sentiment summary
Strengths & Weaknesses	What customers love and complain about
Market Positioning	How the product compares in context
Recommendations	Actionable marketing suggestions

Application Architecture

Single-entry-point Streamlit app — navigation driven by **st.session_state.view** (no multi-page routing, no session state loss).

View	Trigger	Back To
"upload"	Default / Back from results	—
"results"	Analysis completes	"upload"
"summary"	AI Summary button	"results"

File	Responsibility
app.py	Session init, sidebar, single render call
routes/home.py	Reads session_state.view, delegates to view
routes/views/upload.py	Upload/load UI + run_analysis()
routes/views/results.py	Classification metrics, charts, clustering explorer
routes/views/summary.py	Groq summary config form and output
utils/data_processing.py	clean_data(), classify_sentiments(), cluster_categories()
utils/api.py	prepare_groq_context(), create_groq_prompt(), call_groq_api()

Technology Stack

Category	Library / Model	Notes
Sentiment Model	distilbert-base-uncased-finetuned-sst-2-english	HuggingFace Transformers
Embedding Model	all-MiniLM-L6-v2	sentence-transformers
LLM	mixtral-8x7b-32768	Groq API
Data / ML	pandas, numpy, scikit-learn, scipy	Regression, scaling, clustering
Web App	Streamlit	Single-page, session-state nav
Compute	Google Colab + FastAPI	GPU inference via HTTP tunnel
Config	python-dotenv	COLAB_API_URL, GROQ_API_KEY

Results & Evaluation

Stage	Outcome
Classification	Strong on positive class (72% of data). Neutral hardest due to class imbalance. High average confidence scores (>0.85).
Clustering	Cleanly separates Kindle, Fire, Echo, and generic products. Ward linkage produces compact, well-separated groups.
Scoring	$R^2 \sim 0.6\text{--}0.7$. Rating is the strongest predictor. Bayesian ranking meaningfully reorders low-review products.
Summarisation	Mixtral produces coherent professional summaries. Overview and Strengths & Weaknesses most consistent.

Future Work

Completed - Sentiment classification with DistilBERT - Category clustering with MiniLM + Ward linkage - Product scoring with learned weights + Bayesian rank - Groq LLM summarisation - Single-page Streamlit app with view navigation - Colab GPU backend via FastAPI	Planned - Surface product scoring/ranking in results view - Better Groq prompt quality for richer outputs - Automated scheduled report export - Multi-language support - Model retraining pipeline with new data
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Full experimental code, model evaluation, and visualisations are in `overview.ipynb`. See `README.md` for setup and configuration.